Date:10/10/26

CO 1 – ASSESSMENT TOOL -1

APPLICATION BASED QUESTIONS

COURSE CODE: **DSA0308**

COURSE NAME: **NLP**

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Q1. Resume Information Extraction

Aim

To develop a resume information extraction system using Regular Expressions to identify candidate details and check eligibility.

Theory

Resume Information Extraction is the process of collecting important information from resumes automatically. Regular Expressions (Regex) help identify specific patterns in text.

Working

1. Name Extraction: Identifies the candidate's name.
2. Email Extraction: Finds the candidate's email address.
3. Mobile Number Extraction: Identifies the mobile number.
4. Skills Detection: Finds skills such as Python, Java, SQL, Machine Learning, and NLP.
5. Experience Extraction: Identifies the candidate's years of experience.
6. Eligibility Check: Selects candidates with at least 2 years of experience and Python skill.
7. Profile Summary: Displays all extracted details in an organized format.

Example

Candidate details:

1. Name: Arun Kumar
2. Email: arun@gmail.com
3. Skills: Python, Java, SQL
4. Experience: 3 years

Output: Candidate is eligible because they have 3 years of experience and know Python.

Result

Thus, the system extracts candidate information using Regex and identifies eligible candidates.

Q2. Product Search System

Aim

To create a product search system using Regular Expressions to find products based on different keywords.

Theory

A product search system helps users find products easily. Regular Expressions are used to search product names using different methods, such as exact, prefix, suffix, partial, and case-insensitive searches.

Working

1. Exact Search: Finds a product that exactly matches the given keyword. *Example:* Apple iPhone finds Apple iPhone.
2. Prefix Search: Finds products whose names begin with the given keyword. *Example:* Apple finds Apple iPhone and Apple Laptop.
3. Suffix Search: Finds products whose names end with the given keyword. *Example:* Laptop finds Apple Laptop and HP Laptop.
4. Partial Search: Finds products that contain the given keyword anywhere in their names. *Example:* Mobile finds Samsung Mobile.
5. Case-Insensitive Search: Finds products without considering uppercase or lowercase letters.
   Example: apple finds Apple iPhone and Apple Laptop.
6. Search Report: Displays the matching product names and the total number of products found.

Example Output

Search Type	Keyword	Total Products Found
Exact	Apple iPhone	1
Prefix	Apple	2
Suffix	Laptop	2
Partial	Mobile	1
Case-insensitive	apple	2

Result

Thus, the product search system uses Regular Expressions to find products using different search methods and displays the matching products and their total count.

Q3. University Registration System

Aim

To develop a student registration system using Regular Expressions to validate student details.

Theory

A university registration system checks whether student information follows the required format. Regex is used to validate details before course enrollment.

Working

1. Register Number: Checks whether the register number contains the required number of digits.
2. Institutional Email: Checks whether the email follows the university's email format.
3. Course Code: Checks whether the course code follows the required format.
4. Semester: Checks whether the semester is within the accepted range.
5. Mobile Number: Checks whether the mobile number follows the required format.
6. Validation Messages: Displays whether each field is valid or invalid.
7. Final Status: Displays "Registration Successful" if all fields are valid; otherwise, it displays "Registration Failed".

Example Output

Field	Example Value	Status
Register Number	2312345678	Valid
Email	student@university.edu	Valid
Course Code	CSE101	Valid
Semester	Semester 3	Valid
Mobile Number	9876543210	Valid

Final Status: Registration Successful

Result

Thus, the system validates student details using Regex and determines whether the registration is successful.